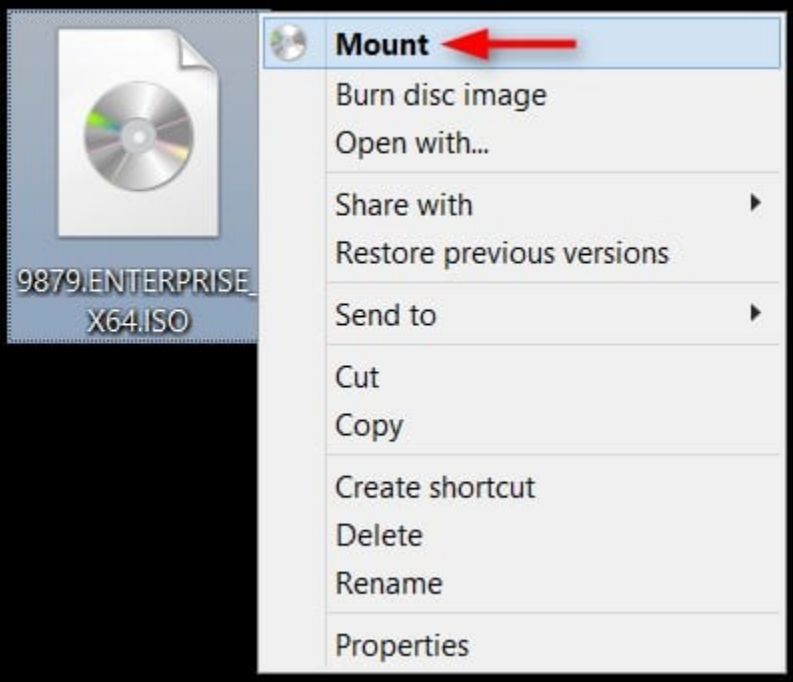
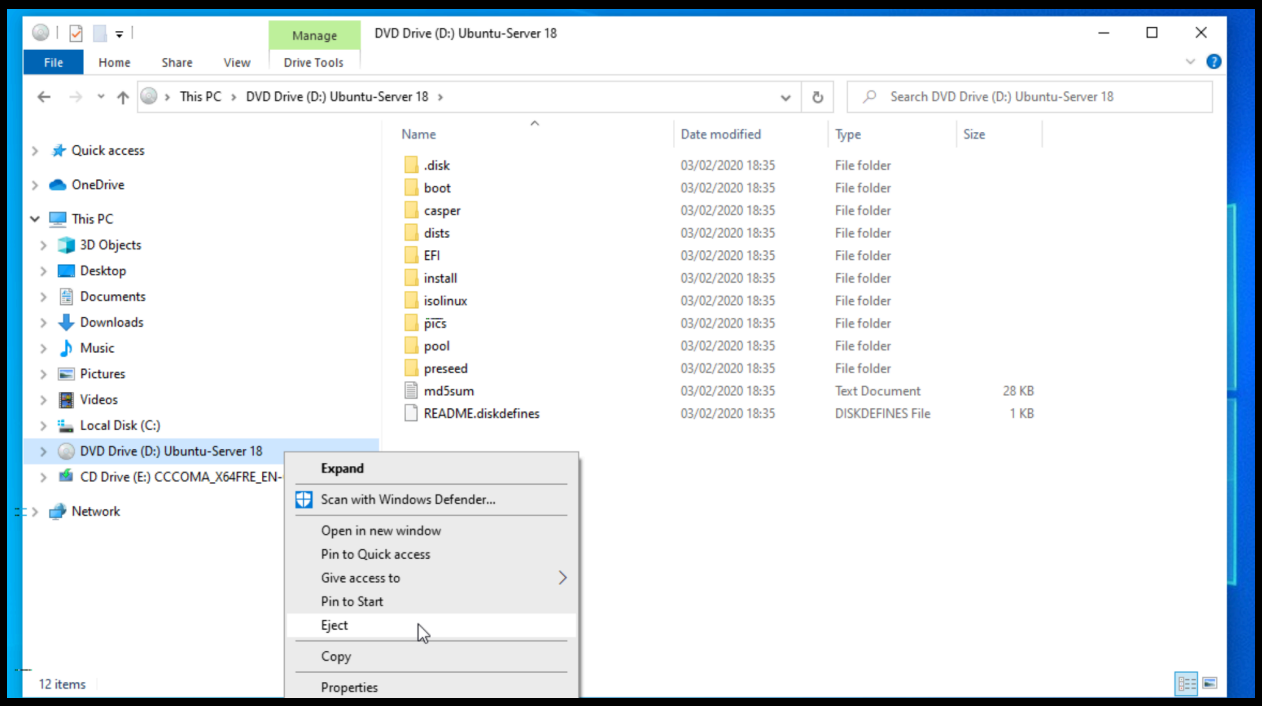
ISO Files and OS Installs



appear as if it is part of a real CD or DVD.
ISO files are archived files containing files and folders organized in a specific manner so as to make them mountable or bootable.

A mountable ISO file is one that can be made to

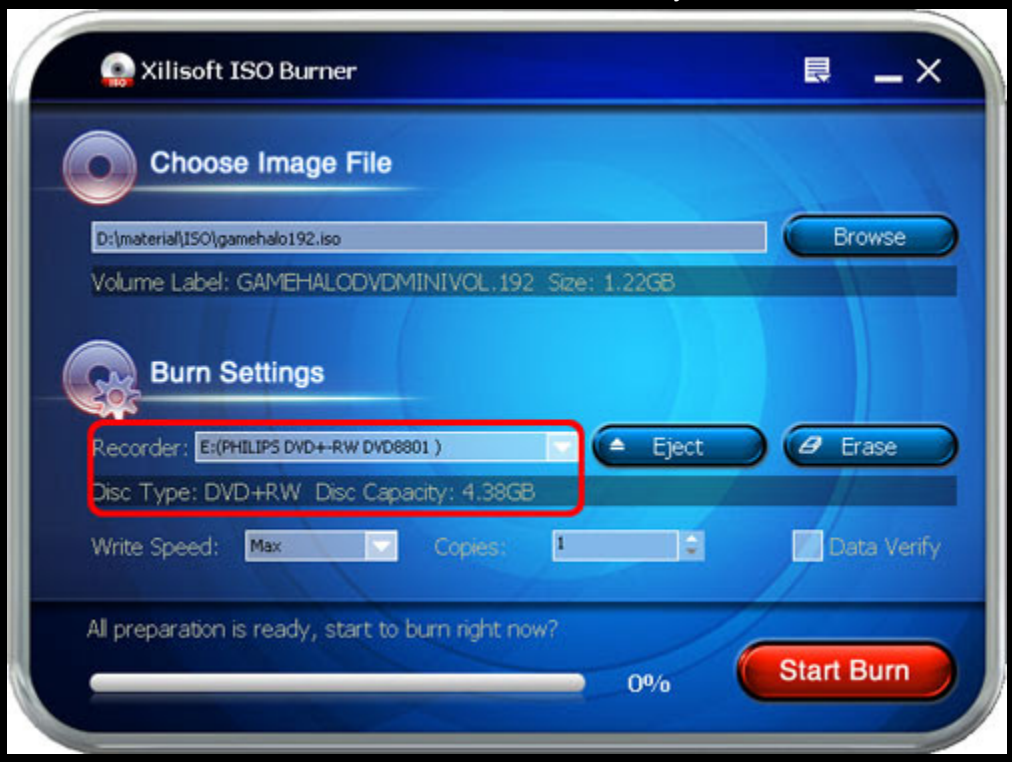


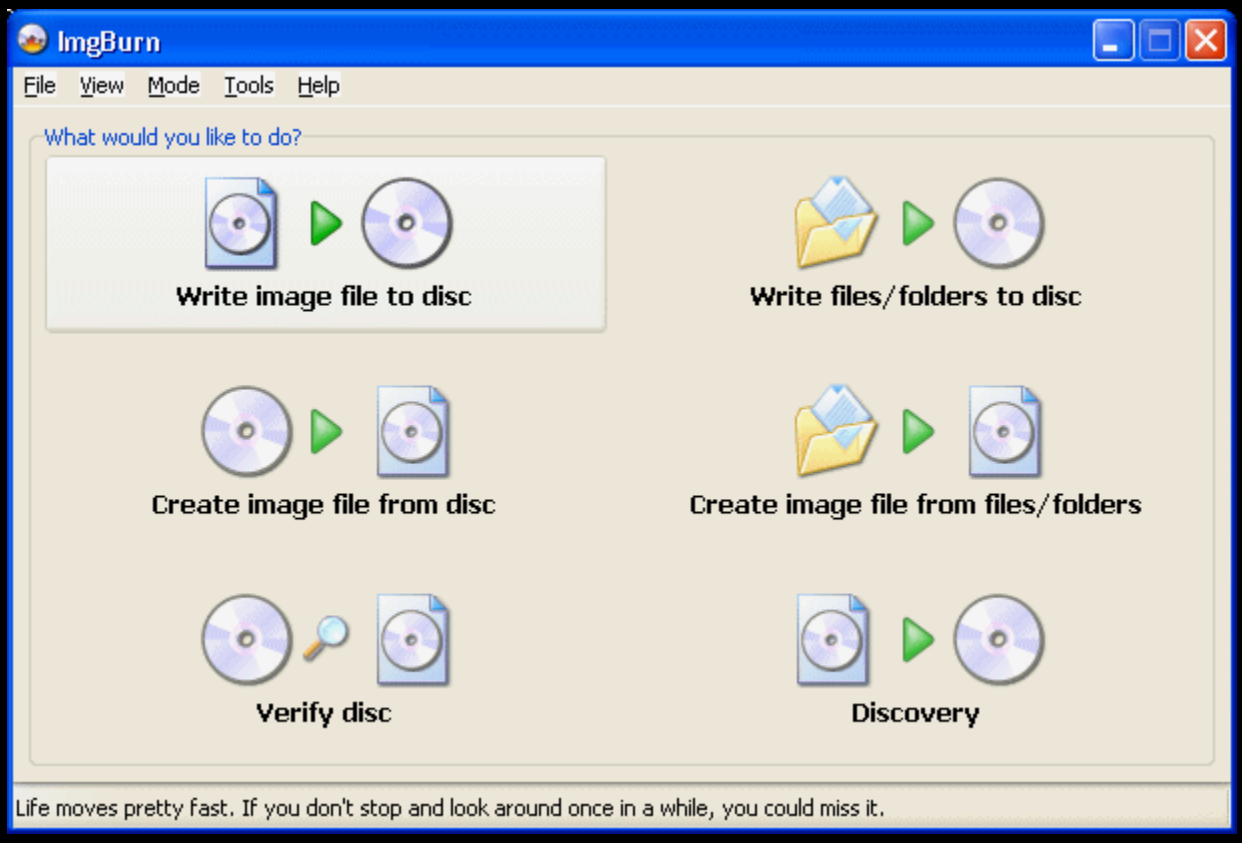


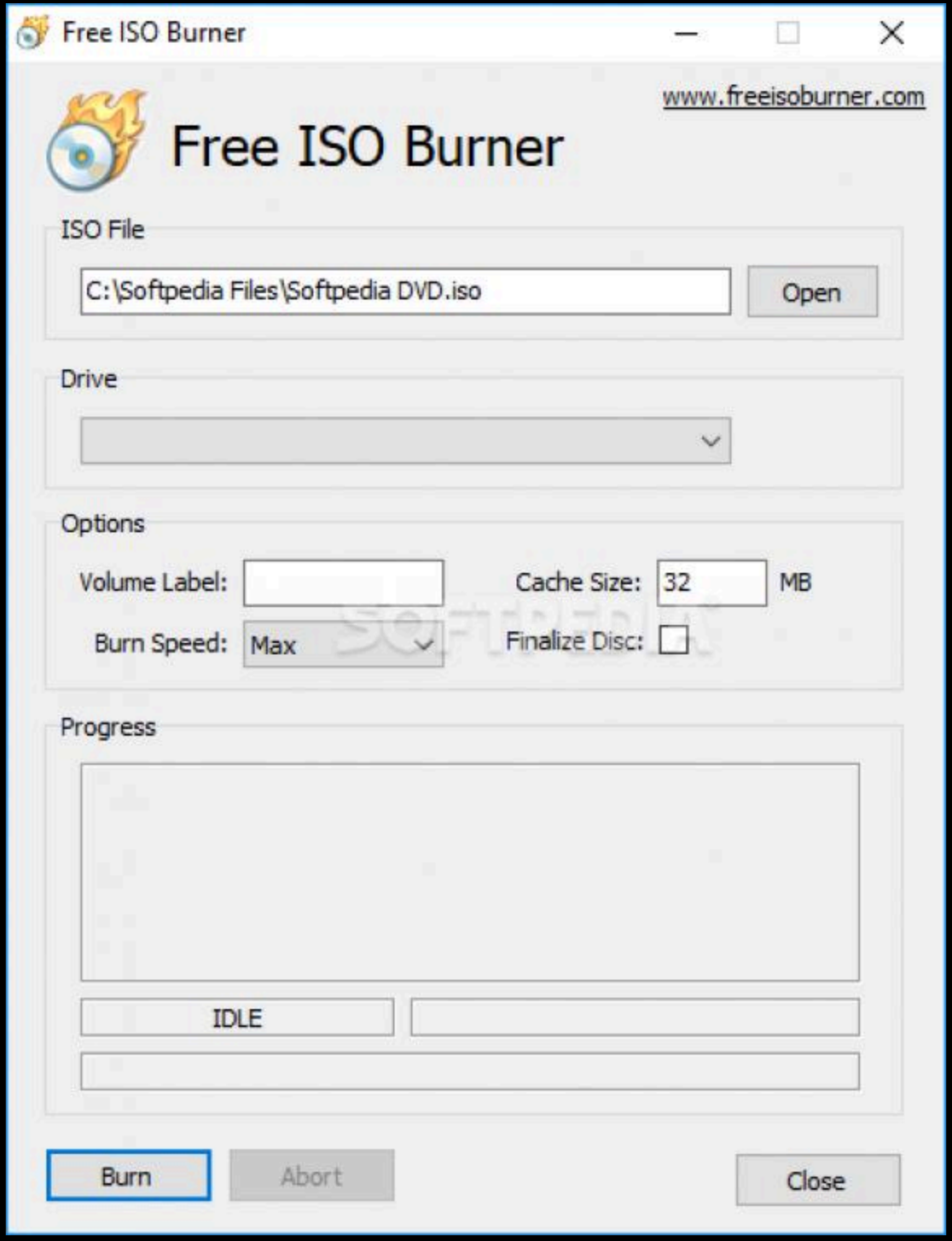
In order to mount an ISO you need software installed or pre-installed in your OS. Mounting an ISO means you will assign a drive letter to it in Windows so that it appears as a real device. Windows 10 and 11 have this software already preinstalled.

A bootable ISO file is one that can be “burned” to a usb or dvd in a format that makes it recognizable by your computer when it boots up. It can take the place of an OS. Actually, most if not all bootable ISO drives are simply operating systems that have been customized with special software i.e. rescuezilla, clonezilla, GParted etc.

To “burn” an ISO to a USB drive you will need specialized software. Some of the best examples of this are Rufus and Balena Etcher. There are many others as well.

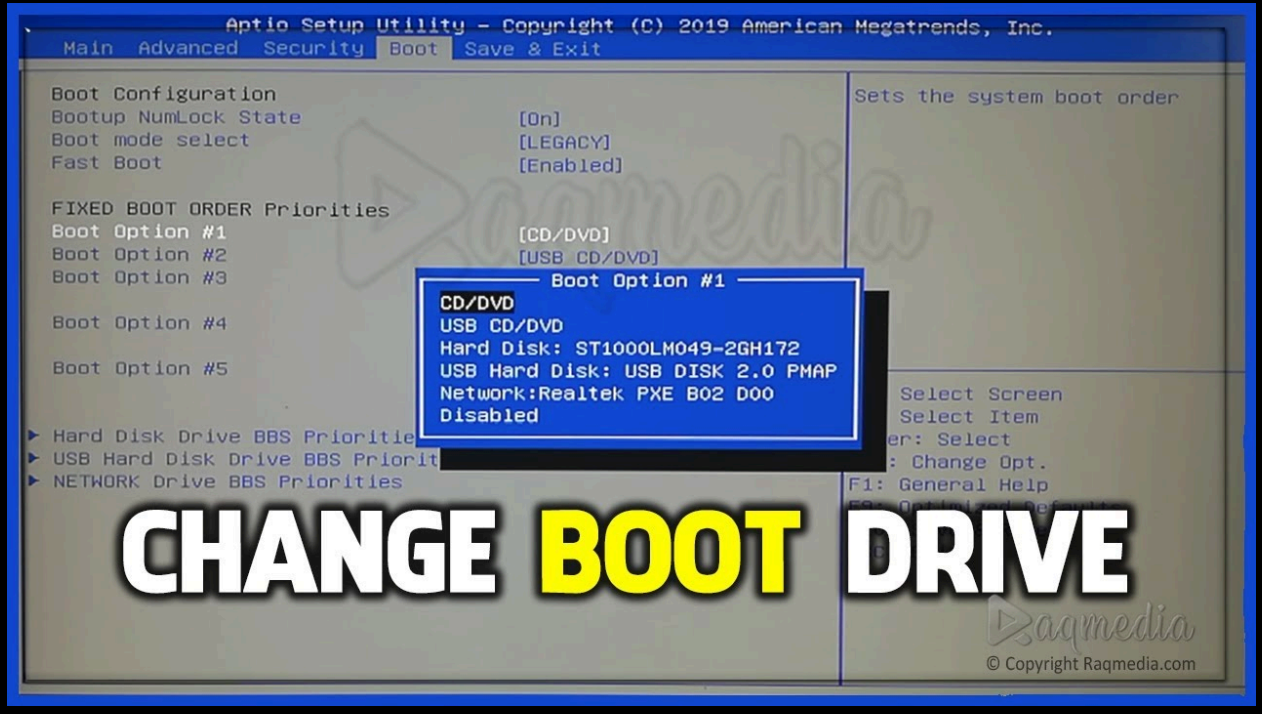


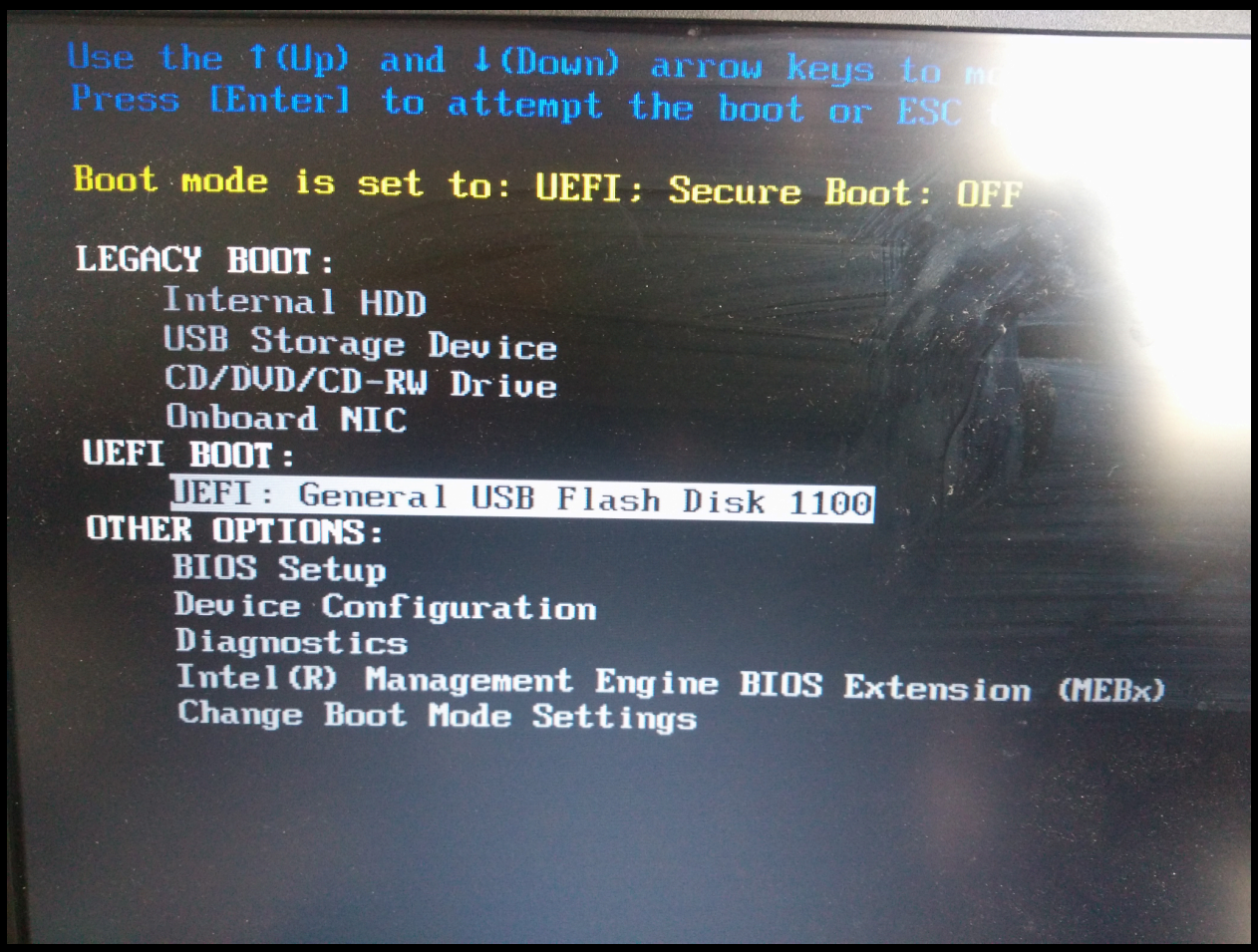




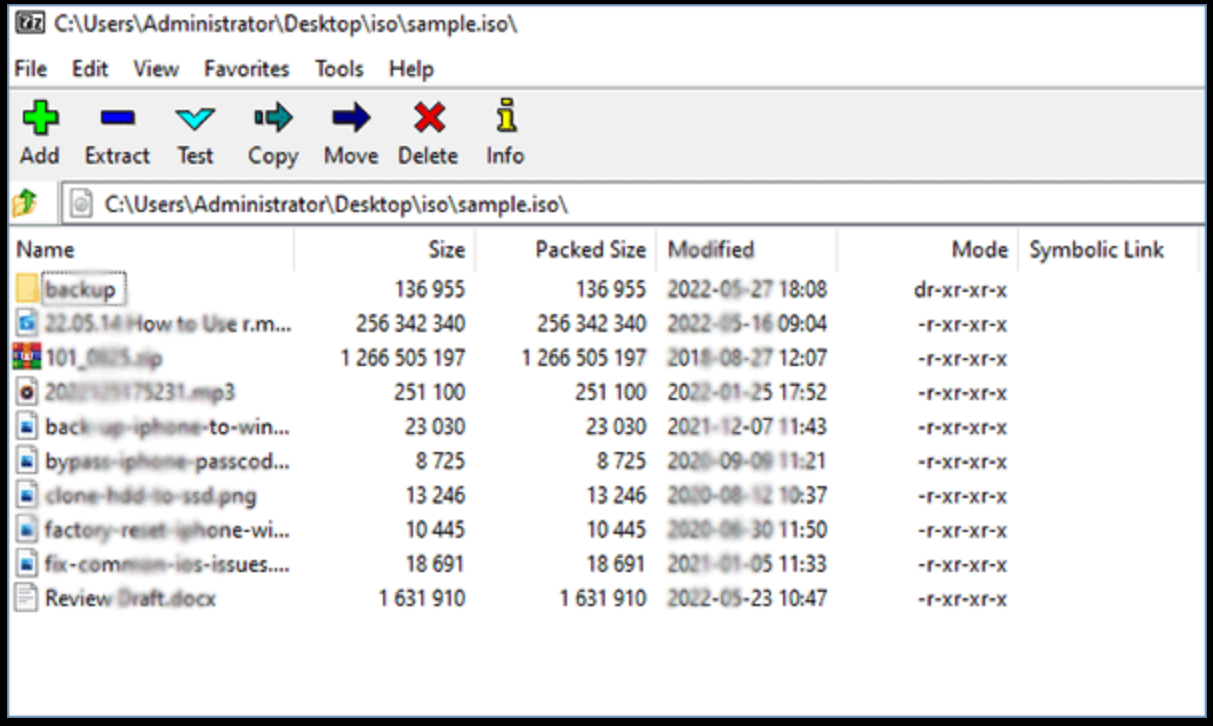
Normally in these “burners” you simply pick the ISO to burn and then tell it where the USB/DVD is for it to burn too. Once completed you should have a bootable USB/DVD.

To boot a USB you will need access to the computer’s BIOS and/or UEFI. The BIOS/UEFI is the software that was installed on a BIOS chip on your motherboard that has the code the computer follows to start up your computer when you power it on. The BIOS/UEFI stores its information/settings in a CMOS chip powered by the CMOS battery. One of the settings you can change is the order of the devices the computer will boot from. Normally it looks at the HDD first to load an OS and then if it doesn’t find one it goes to the next device in the list. To get it to boot from a USB you should enter the BIOS and tell it to boot from the USB first and the HDD second. You can access the BIOS/UEFI by pressing one of the F or function keys (see MB manual).





Since ISO files are archives of other files/folders you can extract them as well. Using a program like WinRAR or 7-Zip will help unpack their contents.



Questions and Exercises

1. What is an ISO file?

   An ISO file (or “ISO image”) is a single file that contains an exact digital copy of an entire optical disc (CD, DVD, Blu-ray). It stores all the data on the disc — files, folder structure, metadata, file system — sector by sector, as a single binary file.

2. What does an ISO burning program do?

   An ISO burning program takes an ISO file and writes (“burns”) it onto a physical medium — e.g., a blank CD, DVD, or Blu-ray disc — recreating the disc as it originally was.

3. What is the purpose of 7-ZIP?

   7-ZIP extracts files and can open ISO files like a folder.

4. What does it mean to mount an ISO file?

   Mounting makes the ISO act like a real disc using a virtual drive.

5. Find a URL to a website offering an ISO of any version of the linux OS. List the URL here.

   Ubuntu: https://ubuntu.com/download

6. Find the URL to the websites that offer Rufus and Balena Etcher. List them here.

   Rufus: https://rufus.ie

   balenaEtcher: https://etcher.balena.io/

7. Find the URLs to at least 2 other sites that have other free ISO burners.

Free ISO Burner: https://www.freeisoburner.com/

ImgBurn: https://www.imgburn.com/

8. Find and list at least 2 URLs to sites offering free ISO mounting software.

   WinCDEmu: https://wincdemu.sysprogs.org/

   Virtual CloneDrive: https://www.elby.ch/en/products/vcd.html

9. Find and list the URLs to 2 youtube videos discussing BIOS and UEFI. Watch them both.

   You may need a flash drive and a portable browser to download the following files. Ask your teacher for one. Your teacher should have provided a link for you to download a portable browser. Once downloaded, put toronto your flash drive and extract it there. Find the program in the extracted folder and run it from the flash drive. You should be able to use this browser to find and download any program back onto your flash drive. Once you have everything on your flash drive you can bring it over to your practice computer to complete the following tasks.

   https://www.youtube.com/watch?v=V4YbWQyq8CU

   https://www.youtube.com/watch?v=YBK3Q7pF1bI

10. *Download the GParted iso at GParted -- Download Get the i686 version (32 bit intel version). Get a copy of 7-ZIP and any ISO burner. Put all three into your practice computer or any available computer running windows. Mount the ISO file (right click on it and select Mount). Find the drive and look at its contents.

    1.Put all three on a USB drive.

    2.Mount the GParted ISO in Windows (right-click → Mount) and check the files.

    3.Use 7-ZIP to extract the ISO contents to a new folder on your desktop.

    4.Burn the ISO to a USB and boot from it to run GParted (Linux disk utility).

    5.Log off and return to Windows.

11. *Using 7-ZIP extract its contents to another folder (create one first) on your desktop. Finally, burn the ISO onto a flash drive and boot the flash drive into GParted. GParted is a linux based utility to manage disks. Log off and boot back into windows.

    URL: https://www.hirensbootcd.org/download/

    Boot from the Hiren ISO.

    Use **ShowKeyPlus** to find your Windows product key and write it down.

12. *Hiren Boot CD is a well known iso utility that many, many IT people have used to help with windows based issues. Download at Download | Hiren's BootCD PE (hirensbootcd.org) and boot using their iso. Using ShowKeyPlus, find your windows product key. Write it down here:

    Boot with Hiren Boot CD.

    Use **Lazesoft Password Recovery** to reset the password.

    Restart and log in with the new password.

    Use Windows to change the password again to **compTech**.

13. *Assume you forgot your windows login password. What can you do? Solution: Boot using the Hiren Boot CD and attempt to reset the password using the Lazesoft Password Recovery tool. Change the password using this tool. Restart your computer and try logging in with your old password. Login with the new password. Once logged in use windows to reset the password again to **compTech**.

    ***1/2 bonus mark each on your last assignment if you demonstrate these four tasks to me for a total of 2 extra marks**